

Lab Assignment No. 3

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Title	Git Feature Branching, Merge Conflict Resolution and Push

Implementation & Theory	Evidence / Description
Check Status & Branches	<pre>PS H:\college\Degree\Sem7\devops\devops-course-work> git status On branch master nothing to commit, working tree clean PS H:\college\Degree\Sem7\devops\devops-course-work> git branch feature-update * master PS H:\college\Degree\Sem7\devops\devops-course-work> </pre> git status and git branch. The repository was on the master branch with a clean working tree, and the existing feature-update branch was listed alongside master.
Create Feature Branch	<pre>PS H:\college\Degree\Sem7\devops\devops-course-work> git checkout -b conflict-feature Switched to a new branch 'conflict-feature' PS H:\college\Degree\Sem7\devops\devops-course-work> git status On branch conflict-feature nothing to commit, working tree clean</pre> git checkout -b conflict-feature. A new branch named conflict-feature was created and switched to. git status confirmed a clean working tree on the new branch.
Commit on Feature Branch	<pre>PS H:\college\Degree\Sem7\devops\devops-course-work> nvim .\README.md PS H:\college\Degree\Sem7\devops\devops-course-work> git add .\README.md PS H:\college\Degree\Sem7\devops\devops-course-work> git commit -m "Updated README on feature branch" [conflict-feature 57186fc] Updated README on feature branch 1 file changed, 1 insertion(+)</pre> README.md was edited using nvim on the conflict-feature branch, staged with git add, and committed with the message "Updated README on feature branch" (commit 57186fc).
Commit on Master Branch	<pre>PS H:\college\Degree\Sem7\devops\devops-course-work> nvim .\README.md PS H:\college\Degree\Sem7\devops\devops-course-work> git add .\README.md PS H:\college\Degree\Sem7\devops\devops-course-work> git commit -m "Updated README on master" [master 130aaf9] Updated README on master 1 file changed, 1 insertion(+)</pre> README.md was independently edited on the master branch using nvim, staged, and committed with the message "Updated README on master" (commit 130aaf9), creating diverging histories.

Lab Assignment No. 3 — Implementation & Theory (Continued)

Implementation & Theory	Evidence / Description
Merge Conflict	<pre>PS H:\college\Degree\Sem7\devops\devops-course-work> git merge conflict-feature Auto-merging README.md CONFLICT (content): Merge conflict in README.md Automatic merge failed; fix conflicts and then commit the result. PS H:\college\Degree\Sem7\devops\devops-course-work> git status On branch master You have unmerged paths. (fix conflicts and run "git commit") (use "git merge --abort" to abort the merge) Unmerged paths: (use "git add <file>..." to mark resolution) both modified: README.md no changes added to commit (use "git add" and/or "git commit -a")</pre> git merge conflict-feature was run from master. Git reported a merge conflict in README.md since both branches modified the same lines, and git status confirmed the unmerged path.
Resolve Conflict & Commit	<pre>PS H:\college\Degree\Sem7\devops\devops-course-work> cat .\README.md # Readme File This is ReadMe File for Devops Course Work Current Status :- Assignment 1 : Added Assignment 2 : Added <<<<<< HEAD Assignment 3 : WIP ===== Assignment 3 : Added >>>>>> conflict-feature PS H:\college\Degree\Sem7\devops\devops-course-work> nvim .\README.md PS H:\college\Degree\Sem7\devops\devops-course-work> cat .\README.md # Readme File This is ReadMe File for Devops Course Work Current Status :- Assignment 1 : Added Assignment 2 : Added Assignment 3 : Added PS H:\college\Degree\Sem7\devops\devops-course-work> git add . PS H:\college\Degree\Sem7\devops\devops-course-work> git commit -m "fixed merge conflict" [master b0aeb97] fixed merge conflict</pre> cat README.md showed the conflict markers (<<<<<<, =====, >>>>>>). The file was edited to keep both updates, re-checked with cat, staged with git add ., and committed with "fixed merge conflict" (commit b0aeb97).
Push to GitHub	<pre>PS H:\college\Degree\Sem7\devops\devops-course-work> git push origin master Enumerating objects: 9, done. Counting objects: 100% (9/9), done. Delta compression using up to 16 threads Compressing objects: 100% (7/7), done. Writing objects: 100% (7/7), 778 bytes 778.00 KiB/s, done. Total 7 (delta 3), reused 0 (delta 0), pack-reused 0 (from 0) remote: Resolving deltas: 100% (3/3), completed with 1 local object. To https://github.com/xzeyo/devops-course-work f2ba8ae..b0aeb97 master -> master</pre> git push origin master pushed the merged, conflict-resolved master branch to the GitHub repository, updating the remote reference from f2ba8ae to b0aeb97.
Conclusion	A feature branch (conflict-feature) was created and updated independently of master, causing a merge conflict in README.md upon merging. The conflict was identified, resolved manually, and committed, and the final merged master branch was pushed to GitHub successfully.
Keywords	status; branch; checkout -b; add; commit; merge; conflict; conflict markers; push.